一般外科外科操作手冊

# Hepatobiliary operation

肝膽外科手術包含種類繁多，例如膽結石、肝內結石、良惡性腫瘤、次發性轉移腫瘤，治療包括切除、膽道截石、T 型管膽汁引流

## Clinical evaluation

1. history（以下列舉常遺漏的項目）
   - oral contraceptives => FNH
   - history/risk factors for liver disease or cirrhosis => family history, tattoo, ear piercing, blood transfusion, smoking, diabetes, ….
   - history/findings of extrahepatic malignancy? => head and neck cancer, breast cancer, ….
   - travel history or other features suggesting an amebic or pyogenic abscess
   - anticoagulant or antiplatelet use?

2. Physical examination
   Please check
   - extrahepatic primary malignancy?
   - Cirrhosis and Portal hypertension?
   - Liver malignancy with metastases?

3. lab data
   - CBC
   - WBC D/C
   - PT, aPTT, and bleeding time if needed
   - liver tests: AST, ALT, Alk-P, Child-Pugh criteria
   - $\alpha$-GT
   - Bil. D/T
   - renal function
   - cholesterol
   - albumin/globulin
   - Virus profile: HBsAg, HBeAg, Anti-HBs, Anti-HBc, Anti-HBe,

Anti-HCV, Anti-delta for Hepatocellular carcinoma
- Tumor markers: AFP, CEA, ….and other suspected target
- ICG test if operation considered for patient with cirrhosis (ICG 藥量為體重的一半, 需開兩張 ICG 檢驗單以比較)

4. examination
- CXR
- EKG if > 40y/o or history
- SMA-12, Na, K, Ca, Sugar,
- Check U/A if Foley catheter indwelling is expected
- Others: cardiac echo, pulmonary function test, etc.

5. Image study for diagnosis and op planning:
- Angiography: SMA + celiac (check Plt, PT, creatinine)
- Tri-phase (dynamic) CT
- CTAP
- ERCP
- MRCP
- MRI
- Blood pool scan

## Pre-op preparation

1. Informed consent: 住院治療計畫書，手術同意書，剩餘檢體同意書，人體試驗同意書，自費同意書等等
2. complete cirrhosis and tumor evaluation
   liver reserve? Need portal vein ligation? Future liver remnant?
3. Prepare blood: PRBC, FFP, PC if thrombocytopenia,
   Cryoprecipitate if needed
4. Empirical antibiotics: clean contaminated wound
   - 一般: 可用可不用 1st generation of cephalosporins (ex. Cefazolin 1gm, 30-60 min before skin incision, 1gm q4h during operation; may extend to 1gm q8h 24hr after operation)
   - high risk: 一定要用(ex. valvular disease, ….)
5. consultation: if needed, consult anesthesiologist, cardiologist, pre-op RT education, etc

## Intraoperation

1. NG tube insertion

2. Foley catheter indwelling
3. Intraoperative sonography

## Post-op care

- monitor vital signs
- I’s and O’s
- Drain amount and character: T-tube, JP drains, penrose, sump drain
- Chest care: incentric spirometry? IPPB?
- Pain control
- Encourage ambulation
- Check Hb, Hct, Plt, PT after op if needed
- Check PT, AST, ALT, and Bil. D/T 2 days after operation
- Signs of liver failure?

# Gallbladder disease

The spectrum of gallbladder disease ranges from asymptomatic cholelithiasis to gallbladder (or biliary) colic, cholecystitis, choledocholithiasis, and cholangitis. **Cholelithiasis** is the presence of gallstones in the gallbladder. **Biliary colic** is pain caused by a stone temporarily obstructing the cystic duct. **Cholecystitis** is inflammation of the gallbladder from obstruction of the cystic duct.
**Choledocholithiasis** is the presence of a stone in the common bile duct. Further complications of gallbladder disease include gallstone pancreatitis, gallstone ileus, biliary cirrhosis, and gallbladder cancer. Cholangitis occurs when a gallstone obstructs the biliary or hepatic ducts, causing inflammation and infection.
Pathophysiology:
Gallstones are rocklike collections of material that form inside the gallbladder. **Cholesterol stones** are most common followed by pigment, then mixed stones. The stones form when there is an imbalance or change in the composition of bile. When the ratio of cholesterol to bile acids or phospholipids is increased, bile becomes supersaturated with cholesterol; it crystallizes and forms a nidus for stone formation. Impaired gallbladder motility, biliary stasis, and bile content predispose people to the formation of gallstones. Sludge may be a step in the formation of stones, or it may occur independently.
Five to fifteen percent of patients with acute cholecystitis present without stones (acalculous cholecystitis). This typically occurs in patients with prolonged illness, such as those with major trauma or with prolonged ICU stays.
**Pigment stones**, which comprise 15% of gallstones, are formed by the crystallization of calcium bilirubinate. Diseases that lead to increased destruction of red blood cells (hemolysis), abnormal metabolism of hemoglobin (cirrhosis), or infections (including parasitic) predispose people to pigment
stones. Black stones are found in people with hemolytic disorders. Brown stones are found in the intrahepatic or extrahepatic duct. They are associated with infection in the gallbladder and commonly are found in people of Asian descent.

## Clinical evaluation

History taking

The clinical stages of cholelithiasis are asymptomatic (the presence of gallstones without symptoms), symptomatic (biliary colic), and complicated (eg, cholecystitis, choledocholithiasis, cholangitis). Most gallstones (60-80%) are asymptomatic.

Classically, biliary colic is described as episodic pain in the right upper quadrant that radiates to the right shoulder or back. It begins postprandially (usually within an hour) and may last from 1-5 hours. It is caused by contraction of the gallbladder (in response to a fatty meal) against an obstructing gallstone (or sludge) in the cystic duct. This leads to increased pressure within the gallbladder and pain. Associated symptoms may include diaphoresis, nausea, and vomiting.

In classic cases, pain is in the right upper quadrant; however, visceral pain and gallbladder wall distention may be only in the epigastric area. Once the peritoneum is irritated, pain localizes into the right upper quadrant. Small stones are more likely to be symptomatic than large stones.

## Physical examination

Since the gallbladder is not inflamed in uncomplicated biliary colic, the pain is poorly localized and visceral in origin.

In acute cholecystitis, although nonspecific, a positive Murphy sign (inspiratory arrest on deep palpation of the right upper quadrant during deep inspiration) is highly suggestive of cholecystitis. Fever is often present, but it may lag behind other signs or symptoms. The presence of fever, persistent tachycardia, hypotension, or jaundice necessitate a search for complications of cholelithiasis, including cholecystitis, cholangitis, pancreatitis, or other systemic causes.

## Laboratory Studies

Laboratory study results are normal in the asymptomatic patient and patients with uncomplicated biliary colic. An elevated white count should raise the suspicion for cholecystitis or other infectious process. Elevated transaminases levels indicate a hepatic process, while hyperbilirubinemia and an elevated alkaline phosphatase level are evidence for common bile duct obstruction. An elevated lipase is indicative of pancreatitis, the most common causes of which are alcohol and gallstones.

## Imaging Studies

Ultrasonography: highly sensitive and specific for gallstones greater than 2 mm. It is less so for microlithiasis or biliary sludge.
Ultrasonography is very useful for diagnosing uncomplicated acute cholecystitis. The sonographic features of acute cholecystitis include gallbladder wall thickening (>5 mm), pericholecystic fluid, gallbladder distention (>5 cm), and a sonographic Murphy sign. The presence of multiple
criteria increases its diagnostic accuracy.
Computerized tomography: can be used in diagnostic challenges or to further characterize complications of gallbladder disease. CT is particularly useful for the detection of intrahepatic stones or recurrent pyogenic cholangitis.

## Treatment

Gallstones found incidentally may be followed until they become symptomatic.
Surgical treatment of asymptomatic gallstones without medically complicating diseases is discouraged. The risk of complications arising from interventions is higher than the risk of symptomatic disease. Approximately 25% of patients with asymptomatic gallstones develop symptoms within 10 years.
Persons with diabetes and women who are pregnant should have close follow-up to determine if they become symptomatic or develop complications.
Patients with risk factors for complications of gallstones may be offered elective cholecystectomy, even with asymptomatic gallstones:
- Cirrhosis , Portal hypertension , Sickle cell disease , Children, - Transplant candidates, Diabetes with minor symptoms
- Patients with a calcified or porcelain gallbladder should consider elective cholecystectomy due to the increased risk of carcinoma (25%).

## Medication

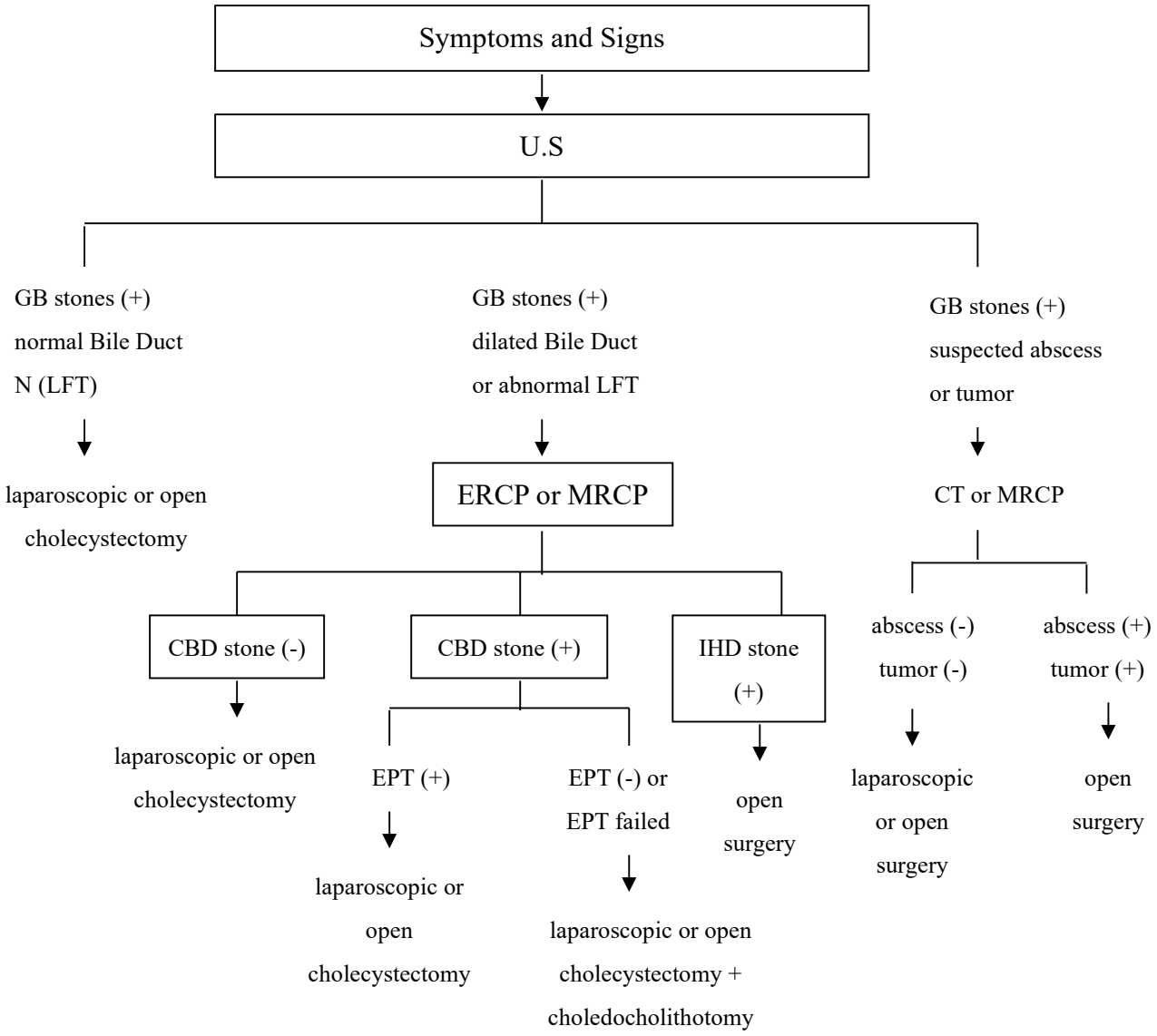
Elective cholecystectomy is the treatment of choice for symptomatic cholelithiasis. In uncomplicated cholelithiasis with biliary colic, medical management may be a useful alternative to cholecystectomy in selected patients, particularly in patients with high surgical risk.

Complications of cholelithiasis may include the following:
- Biliary colic , Cholecystitis (acute or chronic) , Choledocholithiasis , Cholangitis , Sepsis , Gallstone pancreatitis , Gallstone ileus
Prognosis
Less than half of patients with gallstones become symptomatic.
The mortality rate for an elective cholecystectomy is 0.5% with less than 10% morbidity.
The mortality rate for an emergent cholecystectomy is 3-5% with 30-50% morbidity.

Pre-OP preparation:
Prepararion as usual, including op methods and timing
Prophylactic or treatment antibiotics if signs of infection

Postoperative management:
IV fluids
Resume oral intake if bowel sounds are active or have flatus passage, NG decompression if needed
Prophylactic antibiotics for 24 hours or treatment antibiotics if there is contamination at OP orinfection
Drains: record amount and observe for bile leakage

## Guideline for Gall Bladder Stones



# 甲狀腺結節與腫瘤

甲狀腺疾病的外科適應症有 Hyperthyroidism, multinodular goiter, thyroid nodules, and thyroid cancer, 治療包括次全切除、 全葉切除及雙側全葉切除。

## (一)、Pre-operative and Post-operative Care in Thyroid Surgery

1. Pre-operative evaluation
   (1) History and physical examination
   (2) CBC biochemistry, urinalysis, chest X-ray, and EKG
   (3) Thyroid function tests for hyperthyroidism or hypothyroidism
   (4) Thyroid ultrasound study and Needle aspiration of the thyroid

2. Pre-operative care
   Be sure that the patient is in euthyroid status.
   (1) For hyperthyroidism (Grave 's disease, toxic multinodular goiter, and toxic adenoma etc.)
      A) Antithyroid drugs (eg., propylthiouracil, methimazole, carbimazole) are usually given at the outpatient clinic (OPD) to achieve euthyroid status
      B) Lugol 's sol' n, 2 - 3 ml t.i.d. for 7 - 10 days, usually given at OPD and should be continued to the night before operation
      C) Do not use prophylactic antibiotics.
   (2) For non-toxic nodular goiter and other non-toxic thyroid disease
      A) No specific mediation is needed.
      B) Do not give prophylactic antibiotics.

3. Post-operative care
   (1) Continue IV fluid overnight
   (2) Do not give antibiotics
   (3) Oral intake of food and analgesics are allowed.
   (4) Record drain output amount, pay attention to increased output .
   (5) Check serum calcium level p.r.n. Give calcium supplement only if symptoms of hypocalcemia are present
   (6) Tracheostomy set at bed side

(7) If dyspnea occurs and is suspected to be caused by hematoma, open the wound and evacuate the hematoma immediately. **This is a life-saving procedure.**

(8) If thyroid storm (e.g. fever, diaphoresis, tachycardia, arrhythmias, hyperkinetic and somolent etc.) is suspected, start treatment with:

A) Propylthiouracil (PTU) 1000 mg, stat, followed by 200 mg q6h until the patient is euthyroid

B) Propranolol: 0.5 - 2 mg IV q4h or 20 - 80 mg orally q6h

C) Give hydrocortisone 100 - 200 mg daily if adrenal insufficiency is suspected

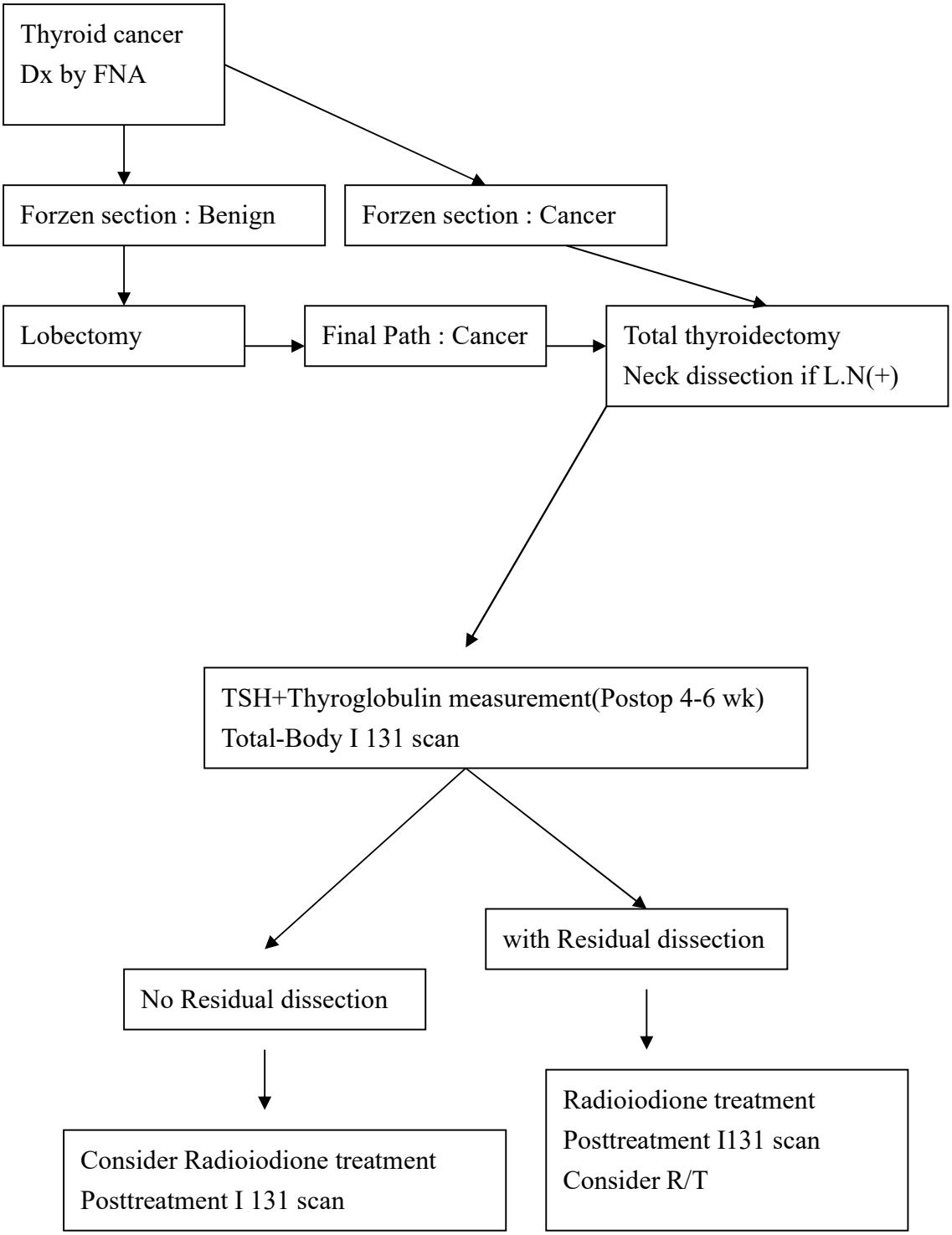
(9) If symptomatic hypocalcemia occurs, give calcium gluconate 1 gm IV push slowly immediately and then 2 gm in IV fluid continuously q8h

(10) If hypocalcemia persists, oral calcium is begun at the dosage of 1.5 - 2.0 gm calcium ion per day (1 gm of elemental calcium daily = 8 gm of calcium lactate)

(11) For parathyroid surgery, iPTH level should be checked postoperatively to confirm a successful operation (a >50% reduction in iPTH level should be observed)

(12) **Ask for help (residents) in condition of (4)~(11) when on duty**

# Management of Thyroid Nodules



# 胃腸外科疾病

## 1.Upper Gastrointestinal (GI) Bleeding

Definition: Bleeding proximal to the ligament of Treitz
Surgical Indication:

(1)Unstable vital signs s/p blood transfusion ( >6U bloods/ day)
(2)Recurrent and active bleeding
(3)Medical treatment failure
(4)Lack of blood preparations

## 2. Peptic Ulcer

Surgical Indications:

1. Perforation/hemorrhage 2. Gastric outlet obstruction 3. Intractable pain

## 3.Gastric cancer

臨床表現以腹痛、 體重下降為主，有些人會有出血或阻塞嘔吐。理學檢查應包括淋巴結探測及肛門檢查。Tumor marker: CA199, CEA, AFP 。影像檢查有 PES +/- Bx，UGI series，Abd CT or US and EUS (Optional)

## 4. GIST (Gastrointestinal tumors)

GIST 臨床表現不明顯，有些人會有出血表現。. 影像檢查 Enahanced Abd CT 及 CxR 決定大部分的治療
If the tumor is resectable with minimal morbidity, surgical resection is indicated
If the tumor is too large or high risk in operation then Neoadjuvant therapy should be considered

## 5. Intestinal Obstruction and appendicitis

均屬腹部急症，須給予點滴輸液治療。當 Intestinal Obstruction 有腹膜炎或敗血症跡象，須準備手術。Appendicitis 以手術為主，但是 tumor formation 則以 antibiotics 優先。

## Pre-operative managements:

(1) Physical examination & history
(2)Place a large bore IC catheter / CVP line and Foley catheter

(3)Prepare and give packed RBC/ FFP if needed
(4)Treat hypovolemic shock: with crystalloid, HAES or L/R challenge
(5)Check Hb, biochemistry, arrange Chest X-ray, EKG
(6)Diagnostic approach:
A) N/G tube insertion with cold N/S irrigation
B) Arrange endoscopy for diagnosis and treatment
C) Angiography - in selective cases
(7)Sign 剩餘檢體同意書 for tumor specimen after explanation was done by surgeons

## Post-operative management:

(1)Check vital signs, intake/output q4h, initially then routine
(2)Watch for NG output for the possibility of rebleeding
(3)IV fluid supply
(4)Recheck hemoglobin (Hb). Give packed RBC if Hb < 10 gm%
(5)Antibiotics are given if there is severe intraabdominal contamination ( Wound class III)
(6)Pain control with pethedine (1mg/Kg)
(7)Check Hb, Hct, Na, K, Cl, Ca, BUN, Cr, stat and cm
(8)May remove NG tube then try oral intake on POD 4 if:
(a) NG output is decreasing with clean content
(b) Flatus/stool passage
(9)Remove drain tube on the POD7 to POD 10 in the gastrectomy cases if there is no evidence of leakage found.
(10) Remove stitches on POD 7

### Pre op order:

OP method: 1. Total gastrectomy 2. Subtotal gastrectomy
OP site: abd
OP would be performed on cm, by Dr.----------
Sign A/O permit
Prepare blood: PRBC, FFP 2U
NPO since MN
IVF: D1/4S 1000+/- (Kcl 20meq + RI 8U if DM present) run as maintain dose
Finger sugar q6h if DM
Medication:

Fleet enema HS
Bring Cefazolin 2pc to OR

*Arrange PFT, 2D echo, consult Anesthologist or CV before op if needed

Cares after operation:

1. Check Drainage tube every day, including color, total amount, If persisting high output of blood like fluid, please notice CR and VS Check the OP note for the position of JP drain
2. NPO with NG decompression, if dislodgement of NG found, discuss to VS or CR first before re-on NG tube
3. Intensive chest care including Tri-flow +/- IPPB+/- Chest care
4. Monitor urine output daily
5. May remove foley tube after DC PCA, encourage ambulation and chest care
6. If malnutrition before op, may add PPN (Kabiven or Fat+ lipovenus ) for nutrition supply
7. Check wound condition especially after POD3, if infection present, pt would have local tenderness, reddish, dirty or pus discharges of wound
8. After POD 5 to 7, with flatus/stool passage, Pt may on diet. Remove NG and then try sip water first, then order PG diet for Pt (mostly PG diet 135 or PG diet 246 in subtotal gastrectomy Pt, PG1235 for total gastrectomy Pt)
9. Tapping IVF and PPN if Pt oral take is well
10. Remove stitches on POD 7 if no wound infection found
11. May remove VB after oral intake well and no leakage found

# Breast Surgery

乳房手術處理良性腫瘤切除及惡性腫瘤治療

1. History taking should include the following risk factors:
   (1)Family history, past history
   (2)Reproductive history: menarche, menopause, menstruation, age of each pregnancy
   (3) Specific medication history

2. Physical examination should include the followings:
   (1) The mass:
   Location(right or left, clock, distance from nipple), size, duration, consistency.
   (2) Skin: reddness, edema, dimple
   (3) Nipple: discharge, bleeding, retraction, eczema
   (4) Axillary nodes, supraclavicular nodes
   (5) Systemic check-up

3. Diagnosis:
   (1) Mammography or echo if available
   (2) Fine needle aspiration cytology or core needle biopsy

4. Pre-operative order:
   (1)Routine examination includes CBC, liver function, ECG, chest PA, CEA,
   CA15.3
   (2) In cases of local advanced cancers, do bone scan and liver echo
   (3) Set up IV on the opposite to surgical site arm
   (4) Prepare packed RBC 2 - 4 units if severe anemia (Hb 7.0) is present.
   (5) No prophylactic antibiotics is needed except in infected cases
   (6) SLN nuclear scan (檢查:核子醫學科: venography) if SLN biopsy is scheduled(前一天先電聯核醫科)

5. Clinical staging:  TNM staging

6. Surgery

(1)Breast conservative surgery may be performed for carcinoma in situ or stage I tumors.

(2) Modified radical mastectomy (Patey type) is the conventional standard procedure for operable cases.

(3) Radical mastectomy (Halsted) is indicated if tumor involves pectoralis muscles.

(4) Breast conservative surgery and sentinel LN biopsy is an alternative to MRM; if frozen section proved positive sentinel LN, axillary LN dissection should be performed,

7. Chemotherapy: following the treatment protocol

8. Radiotherapy

(1)After conservative surgery

(2)For local control - to the internal mammary node, supraclavicular node, which is possibly involved

(3) For locally advanced cancers

(4) As primary treatment

9. Hormonal therapy

(1)For menopausal, premenopausal and perimenopausal patients with positive estrogen receptor: Tamoxifen 20 mg daily

(2)Surgical ablative procedure of endocrine organs in metastatic cases: Oophorectomy, adrenalectomy, hypophysectomy

10. Reconstruction

11. Psychosocial rehabilitation

12. Post-operative care and nursing care

(1) Chest care

(2) Dressing change, usually on the $3^{rd}$~ $5^{th}$ day after surgery

(3) Removed drain tubes if the amount of discharge is < 30 ml/day; may take home if large amount is recorded

(4) Extremity exercise as early as possible

(5) Give analgesics, but no antibiotics is needed except for contaminated wound